Hail Song

PhD Student · KAIST UVR Lab

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Education

Korea Advanced Institute of Science and Technology *Daejeon, Republic of Korea*
Ph.D. Student, Graduate School of Culture Technology *Mar 2024 - present*
- Advisor: Prof. Woontack Woo

Korea Advanced Institute of Science and Technology *Daejeon, Republic of Korea*
M.S., Graduate School of Culture Technology *Mar 2022 - Feb 2024*
- Advisor: Prof. Woontack Woo
- Thesis : Real-time Cumulative SMPL-based Avatar Body Generation System

Yonsei University *Seoul, Republic of Korea*
B.S., Mechanical Engineering *Mar 2015 - Feb 2021*

Experiences

Aug 2024 - Feb 2025	**Visiting Scholar at Carnegie Mellon University**, Sponsored by Institute of Information & Communications Technology Planning & Evaluation, IITP.
Jul 2023 - Aug 2023	**Visiting Scholar at TUM (Technical University of Munich)**, TUM School of Computation, Information and Technology, CAMP
Mar 2022 - present	**Research Assitant**, Development of Untact Realistic OpenXR Platform Technology. National Research Project funded by Korean Government. Joint Research Project with KISTI, KAIST, KICT, and KIOM.
May 2020 - Feb 2022	**Computer Vision Researcher (Full-time)**, InBody Co., Ltd.
Oct 2016 - Jul 2018	**Mandatory Military Service**, ROK Army sergeant, honorable discharge

Research Interests

Avatar Creation with Neural Rendering

I am dedicated to researching efficient 3D avatar generation. My work focuses on creating high-fidelity avatars from minimal user input, as demonstrated in my research on head avatar reconstruction from a single image using Diffusion and 3DGS [Under review], and real-time full-body avatar generation from monocular RGB-D video [ISMAR 2023]. More recently, I have also explored generating 3D face models from 3D sketches [CVPRW 2025]. My broader interests lie in the diverse applications of neural rendering and advancing 3D human model representations. Ultimately, my goal is to enhance the accessibility of this technology, enabling all users to freely and equitably express their identities in the metaverse.

Avatar Interaction on XR Devices

Recent advancements in neural rendering, including NeRF and 3D Gaussian Splatting, enable photo-realistic avatar creation. However, deploying these on consumer HMDs like the Meta Quest or Apple Vision Pro presents significant challenges. Key obstacles include maintaining novel-view consistency, tracking facial expressions with internal sensors, and estimating body pose from egocentric-only data, all within limited computational budgets. To address these issues, my current research focuses on developing an integrated metaverse communication system that delivers seamless and realistic avatar interactions on XR devices.

Publications

VRGaussianAvatar: Integrating 3D Gaussian Avatars into VR
Hail Song*, Boram Yoon, Seokhwan Yang, Seoyoung Kang, Hyunwoo Kim, Hannes Metzmacher, Woontack Woo
Transactions on Visualization and Computer Graphics (**TVCG**) (Will be presented at IEEE VR 2026)

OFERA: Blendshape-driven 3D Gaussian Control for Occluded Facial Expression to Realistic Avatars in VR
Seokhwan Yang, Boram Yoon, Seoyoung Kang, **Hail Song**, Woontack Woo
Transactions on Visualization and Computer Graphics (**TVCG**) (Will be presented at IEEE VR 2026)

Streamlined Facial Data Collection based on Utterance and Emotional Data for Human-to-Avatar Reconstruction
Seoyoung Kang, Seokhwan Yang, **Hail Song**, Boram Yoon, Jaehoon Kim, Kangsoo Kim, Woontack Woo
Transactions on Visualization and Computer Graphics (**TVCG**) (Will be presented at IEEE VR 2026)

SceneLinker: Compositional 3D Scene Generation via Semantic Scene Graph from RGB Sequences
Seok-Young Kim, Donghyeon Kim, Woojin Cho, **Hail Song**, Seoyoung Kang, Woontack Woo
Transactions on Visualization and Computer Graphics (**TVCG**) (Will be presented at IEEE VR 2026)

HMD-only Controllable 3D Gaussian Avatars in VR: Face and Full-body Demonstration
Seokhwan Yang, **Hail Song**, Woontack Woo
2026 IEEE Virtual Reality Workshop (Research Demo) (**VRW 2026**)

VR Zen Garden: Designing a Virtual Environment for Stress Relief
Hail Song, Jinseok Hong, Seonji Kim, Kyung Taek Oh, Woontack Woo, Sungyoung Kim
2026 IEEE Virtual Reality Workshop (Research Demo) (**VRW 2026**)

Fast Texture Transfer for XR Avatars via Barycentric UV Conversion
Hail Song, Seokhwan Yang, Woontack Woo
2025 IEEE International Symposium on Mixed and Augmented Reality (**ISMAR-Adjunct 2025**)

S3D: Sketch-Driven 3D Model Generation
Hail Song*, Wonsik Shin*, Naeun Lee, Soomin Chung, Nojun Kwak, Woontack Woo
GMCV Workshop at CVPR'25 (**CVPRW 2025**)

The Influence of Emotion-based Prioritized Facial Expressions on Social Presence in Avatar-mediated Remote Communication
Seoyoung Kang, **Hail Song**, Boram Yoon, Kangsoo Kim, Woontack Woo
2024 IEEE International Symposium on Mixed and Augmented Reality (**ISMAR 2024**)

Toward Realistic 3D Avatar Generation with Dynamic 3D Gaussian Splatting for AR/VR Communication
Hail Song
2024 IEEE Virtual Reality Workshop (Doctoral Consortium) (**VRW 2024**)
Best DC Paper Awards

Meta-Rodin: a Multi-Sensory Immersive Sculpting System to Cultivate a Memorable Experience in a Meta-Museum
Sihyun Jeong, Coulon Jina Jung, **Hail Song**, Woontack Woo
Korea Computer Congress 2024 (**KCC 2024**)
Outstanding Presented Paper Award

RC-SMPL : Real-time Cumulative SMPL-based Avatar Body Generation
Hail Song, Boram Yoon, Woojin Cho, Woontack Woo
2023 IEEE International Symposium on Mixed and Augmented Reality (**ISMAR 2023**)

Quantitative Optimal Learning Conditions for Gaussian Splatting Based On Input Quantity and Scene Dynamics
Hyungwoo Jin, Eunhee Jung, **Hail Song**, Woontack Woo
Korea Software Congress 2023 (**KSC 2023**)

Effects of Different Facial Blendshape Combinations on Social Presence for Avatar-mediated Mixed Reality Remote Communication
Seoyoung Kang, **Hail Song**, Boram Yoon, Kangsoo Kim, Woontack Woo
2023 IEEE International Symposium on Mixed and Augmented Reality (**ISMAR-Adjunct 2023**)

Real-Time Music Tracking System in Wearable AR Display for Music Performance
Jiyun Park, **Hail Song**, Seungmin Lee, Juhan Nam, Woontack Woo
2023 The HCI Society of Korea (**HCI Korea 2023**)

Awards, Fellowships, & Grants

2024 **IEEE VR 2024 Best DC Paper Award**, IEEE VR 2024
2024 **KCC Outstanding Presented Paper Award**, KCC 2024
2020 **Department Honors**, Yonsei University
2015 **National Science & Technology Scholarship**, KOSAF (Korea Student Aid Foundation) *$ 26,000*

Teaching Experience

2025 **CTP445 Augmented Reality**, Teaching Assistant *KAIST, Daejeon*
2025 **GCT600/MV600 Augmented Reality Project**, Teaching Assistant *KAIST, Daejeon*
2019 **Teaching Assistant (Coding)**, Samsung Dream Class, supported by Samsung Co., Ltd., *Incheon*

Academic Activities

Invited Talks

2023 Korea Software Congress (KSC), Top-Conference Session

Reviewer

IEEE ISMAR 2024, 2025 / IEEE VR 2026 / Korea Computer Congress (KCC) 2024, 2025